In [4]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

In [5]:
```
df = pd.read_csv("HR DATA_Excel 2.csv")

print("Dataset loaded successfully!")
```

```
Dataset loaded successfully!
```

In [6]:
```
df.head()
```

Out[6]:

	Attrition	Business Travel	CF_age band	CF_attrition label	Department	Education Field	emp no
0	Yes	Travel_Rarely	35 - 44	Ex-Employees	Sales	Life Sciences	STAFF-1
1	No	Travel_Frequently	45 - 54	Current Employees	R&D	Life Sciences	STAFF-2
2	Yes	Travel_Rarely	35 - 44	Ex-Employees	R&D	Other	STAFF-4
3	No	Travel_Frequently	25 - 34	Current Employees	R&D	Life Sciences	STAFF-5
4	No	Travel_Rarely	25 - 34	Current Employees	R&D	Medical	STAFF-7

5 rows × 44 columns

In [7]:
```
df.tail()
```

Out[7]:

	Attrition	Business Travel	CF_age band	CF_attrition label	Department	Education Field	er
1465	Yes	Non-Travel	25 - 34	Ex-Employees	R&D	Technical Degree	STA 19
1466	Yes	Travel_Frequently	25 - 34	Ex-Employees	R&D	Life Sciences	STA 18
1467	Yes	Travel_Frequently	35 - 44	Ex-Employees	Sales	Other	STA 16
1468	Yes	Travel_Rarely	Under 25	Ex-Employees	R&D	Life Sciences	STA 18
1469	Yes	Travel_Rarely	Under 25	Ex-Employees	Sales	Life Sciences	STA 17

5 rows × 44 columns

Check the dataset

In [9]:
```
print("Dataset shape:", df.shape)

print("\nColumn names:")
print(df.columns.tolist())

print("\nData types:")
print(df.dtypes)

print("\nMissing values:")
print(df.isnull().sum())
```

```
Dataset shape: (1470, 44)

Column names:
['Attrition', 'Business Travel', 'CF_age band', 'CF_attrition label', 'Dep
artment', 'Education Field', 'emp no', 'Employee Number', 'Gender', 'Job R
ole', 'Marital Status', 'Over Time', 'Over18', 'Training Times Last Year',
'-2', '0', 'Age', 'CF_attrition count', 'CF_attrition counts', 'CF_attriti
on rate', 'CF_current Employee', 'Daily Rate', 'Distance From Home', 'Educ
ation', 'Employee Count', 'Environment Satisfaction', 'Hourly Rate', 'Job
Involvement', 'Job Level', 'Job Satisfaction', 'Monthly Income', 'Monthly
Rate', 'Num Companies Worked', 'Percent Salary Hike', 'Performance Ratin
g', 'Relationship Satisfaction', 'Standard Hours', 'Stock Option Level',
'Total Working Years', 'Work Life Balance', 'Years At Company', 'Years In
Current Role', 'Years Since Last Promotion', 'Years With Curr Manager']

Data types:
Attrition                    object
Business Travel              object
CF_age band                  object
CF_attrition label           object
Department                   object
Education Field              object
emp no                       object
Employee Number               int64
Gender                       object
Job Role                     object
Marital Status               object
Over Time                    object
Over18                       object
Training Times Last Year      int64
-2                            int64
0                             int64
Age                           int64
CF_attrition count            int64
CF_attrition counts         float64
CF_attrition rate             int64
CF_current Employee           int64
Daily Rate                    int64
Distance From Home            int64
Education                    object
Employee Count                int64
Environment Satisfaction      int64
Hourly Rate                   int64
Job Involvement               int64
Job Level                     int64
Job Satisfaction              int64
Monthly Income                int64
Monthly Rate                  int64
Num Companies Worked          int64
Percent Salary Hike           int64
Performance Rating            int64
Relationship Satisfaction     int64
Standard Hours                int64
Stock Option Level            int64
Total Working Years           int64
Work Life Balance             int64
Years At Company              int64
Years In Current Role         int64
Years Since Last Promotion    int64
Years With Curr Manager       int64
```

```
dtype: object

Missing values:
Attrition                          0
Business Travel                    0
CF_age band                        0
CF_attrition label                 0
Department                         0
Education Field                    0
emp no                             0
Employee Number                    0
Gender                             0
Job Role                           0
Marital Status                     0
Over Time                          0
Over18                             0
Training Times Last Year           0
-2                                 0
0                                  0
Age                                0
CF_attrition count                 0
CF_attrition counts             1233
CF_attrition rate                  0
CF_current Employee                0
Daily Rate                         0
Distance From Home                 0
Education                          0
Employee Count                     0
Environment Satisfaction           0
Hourly Rate                        0
Job Involvement                    0
Job Level                          0
Job Satisfaction                   0
Monthly Income                     0
Monthly Rate                       0
Num Companies Worked               0
Percent Salary Hike                0
Performance Rating                 0
Relationship Satisfaction          0
Standard Hours                     0
Stock Option Level                 0
Total Working Years                0
Work Life Balance                  0
Years At Company                   0
Years In Current Role              0
Years Since Last Promotion         0
Years With Curr Manager            0
dtype: int64
```

In [10]:
```
df.describe()
```

Out[10]:

	Employee Number	Training Times Last Year	-2	0	Age	CF_attrition count	CF_attri cou
count	1470.000000	1470.000000	1470.0	1470.0	1470.000000	1470.000000	2
mean	1024.865306	2.799320	-2.0	0.0	36.923810	0.161224	
std	602.024335	1.289271	0.0	0.0	9.135373	0.367863	
min	1.000000	0.000000	-2.0	0.0	18.000000	0.000000	
25%	491.250000	2.000000	-2.0	0.0	30.000000	0.000000	
50%	1020.500000	3.000000	-2.0	0.0	36.000000	0.000000	
75%	1555.750000	3.000000	-2.0	0.0	43.000000	0.000000	
max	2068.000000	6.000000	-2.0	0.0	60.000000	1.000000	

8 rows × 31 columns

In [13]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 44 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Attrition                   1470 non-null   object
 1   Business Travel             1470 non-null   object
 2   CF_age band                 1470 non-null   object
 3   CF_attrition label          1470 non-null   object
 4   Department                  1470 non-null   object
 5   Education Field             1470 non-null   object
 6   emp no                      1470 non-null   object
 7   Employee Number             1470 non-null   int64
 8   Gender                      1470 non-null   object
 9   Job Role                    1470 non-null   object
 10  Marital Status              1470 non-null   object
 11  Over Time                   1470 non-null   object
 12  Over18                      1470 non-null   object
 13  Training Times Last Year    1470 non-null   int64
 14  -2                          1470 non-null   int64
 15  0                           1470 non-null   int64
 16  Age                         1470 non-null   int64
 17  CF_attrition count          1470 non-null   int64
 18  CF_attrition counts         237 non-null    float64
 19  CF_attrition rate           1470 non-null   int64
 20  CF_current Employee         1470 non-null   int64
 21  Daily Rate                  1470 non-null   int64
 22  Distance From Home          1470 non-null   int64
 23  Education                   1470 non-null   object
 24  Employee Count              1470 non-null   int64
 25  Environment Satisfaction    1470 non-null   int64
 26  Hourly Rate                 1470 non-null   int64
 27  Job Involvement             1470 non-null   int64
 28  Job Level                   1470 non-null   int64
 29  Job Satisfaction            1470 non-null   int64
 30  Monthly Income              1470 non-null   int64
 31  Monthly Rate                1470 non-null   int64
 32  Num Companies Worked        1470 non-null   int64
 33  Percent Salary Hike         1470 non-null   int64
 34  Performance Rating          1470 non-null   int64
 35  Relationship Satisfaction   1470 non-null   int64
 36  Standard Hours              1470 non-null   int64
 37  Stock Option Level          1470 non-null   int64
 38  Total Working Years         1470 non-null   int64
 39  Work Life Balance           1470 non-null   int64
 40  Years At Company            1470 non-null   int64
 41  Years In Current Role       1470 non-null   int64
 42  Years Since Last Promotion  1470 non-null   int64
 43  Years With Curr Manager     1470 non-null   int64
dtypes: float64(1), int64(30), object(13)
memory usage: 505.4+ KB
```

Understand the data

In [14]:
```
print("Attrition count:")
print(df["Attrition"].value_counts())

print("\nAttrition percentage:")
```

```
print(
    df["Attrition"]
    .value_counts(normalize=True)
    .mul(100)
    .round(2)
)
```

```
Attrition count:
Attrition
No     1233
Yes     237
Name: count, dtype: int64

Attrition percentage:
Attrition
No     83.88
Yes    16.12
Name: proportion, dtype: float64
```

Total employees,

Employees left,

Attrition rate

In [15]:
```
total_employees = len(df)

employees_left = np.sum(df["Attrition"] == "Yes")

attrition_rate = np.divide(
    employees_left,
    total_employees
) * 100

print("Total employees:", total_employees)
print("Employees left:", employees_left)
print("Attrition rate:", round(attrition_rate, 2), "%")
```

```
Total employees: 1470
Employees left: 237
Attrition rate: 16.12 %
```

Clean the dataset

In [16]:
```
print("Missing values:")
print(df.isnull().sum().sort_values(ascending=False).head(15))

print("\nDuplicate rows:", df.duplicated().sum())
```

```
Missing values:
CF_attrition counts          1233
Attrition                       0
Education                       0
Environment Satisfaction        0
Hourly Rate                     0
Job Involvement                 0
Job Level                       0
Job Satisfaction                0
Monthly Income                  0
Monthly Rate                    0
Num Companies Worked            0
Percent Salary Hike             0
Performance Rating              0
Relationship Satisfaction       0
Standard Hours                  0
dtype: int64

Duplicate rows: 0
```

In [17]:
```python
hr = df.copy()

# Remove exact duplicate rows, if any
hr = hr.drop_duplicates()

# Check missing values in important analysis columns
important_columns = [
    "Attrition",
    "Department",
    "Gender",
    "Age",
    "Monthly Income",
    "Job Role",
    "Over Time",
    "Job Satisfaction"
]

print(hr[important_columns].isnull().sum())
```

```
Attrition           0
Department          0
Gender              0
Age                 0
Monthly Income      0
Job Role            0
Over Time           0
Job Satisfaction    0
dtype: int64
```

Analyze attrition by department

In [18]:
```python
department_analysis = hr.groupby("Department").agg(
    Total_Employees=("Attrition", "size"),
    Employees_Left=("Attrition", lambda x: (x == "Yes").sum())
)

department_analysis["Attrition_Rate"] = (
    department_analysis["Employees_Left"]
    / department_analysis["Total_Employees"] * 100
```

```
).round(2)

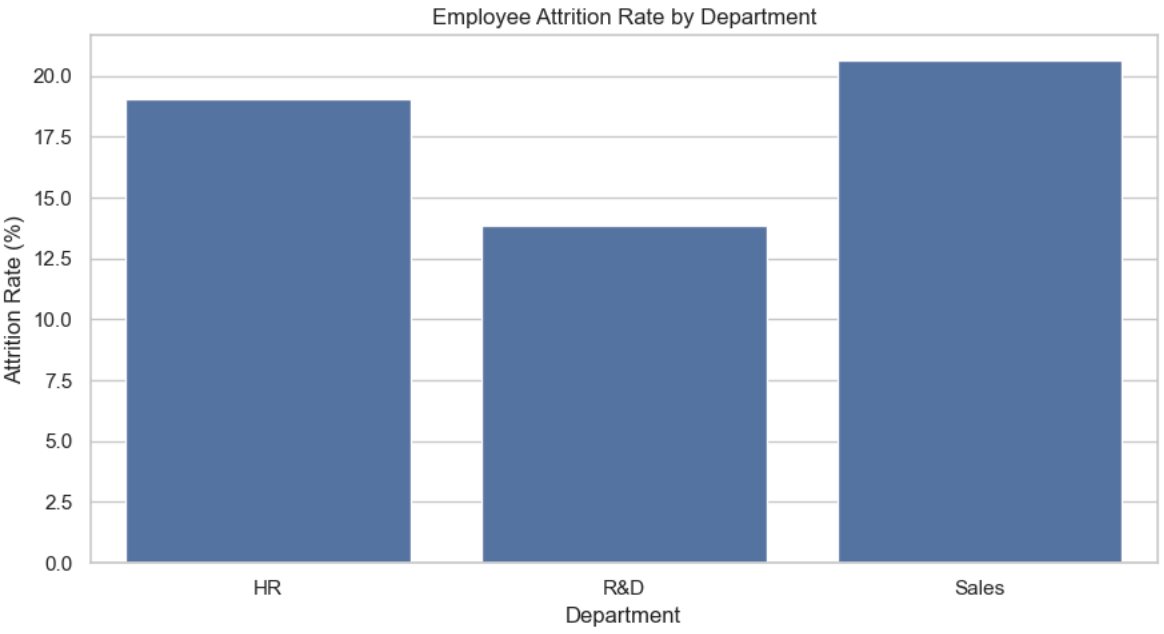
display(department_analysis.sort_values(
    "Attrition_Rate", ascending=False
))
```

Department	Total_Employees	Employees_Left	Attrition_Rate
Sales	446	92	20.63
HR	63	12	19.05
R&D	961	133	13.84

In [19]:
```
plt.figure(figsize=(9, 5))

sns.barplot(
    data=department_analysis.reset_index(),
    x="Department",
    y="Attrition_Rate"
)

plt.title("Employee Attrition Rate by Department")
plt.xlabel("Department")
plt.ylabel("Attrition Rate (%)")
plt.tight_layout()
plt.show()
```



Analyze attrition by overtime

In [20]:
```
overtime_analysis = hr.groupby("Over Time").agg(
    Total_Employees=("Attrition", "size"),
    Employees_Left=("Attrition", lambda x: (x == "Yes").sum())
)

overtime_analysis["Attrition_Rate"] = (
```

```
    overtime_analysis["Employees_Left"]
    / overtime_analysis["Total_Employees"] * 100
).round(2)

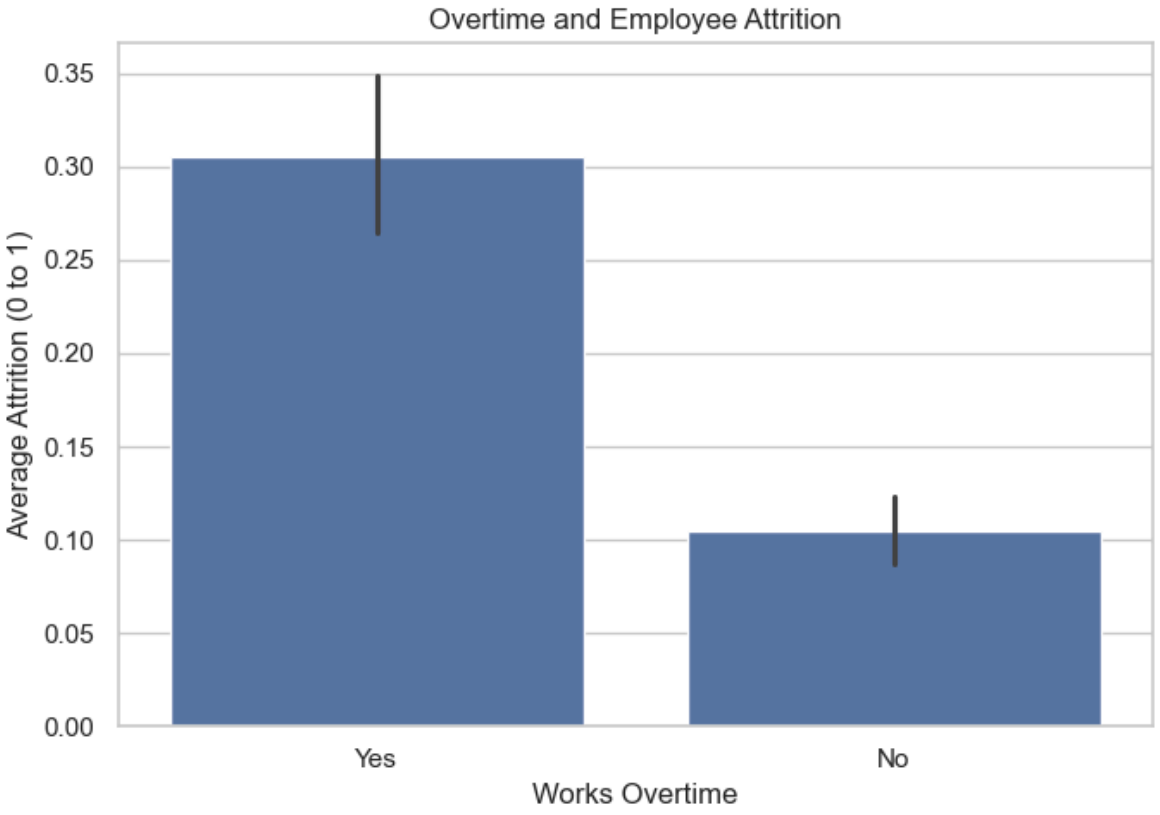
display(overtime_analysis)
```

Over Time	Total_Employees	Employees_Left	Attrition_Rate
No	1054	110	10.44
Yes	416	127	30.53

In [21]:
```
plt.figure(figsize=(7, 5))

sns.barplot(
    data=hr,
    x="Over Time",
    y=hr["Attrition"].eq("Yes").astype(int)
)

plt.title("Overtime and Employee Attrition")
plt.xlabel("Works Overtime")
plt.ylabel("Average Attrition (0 to 1)")
plt.tight_layout()
plt.show()
```

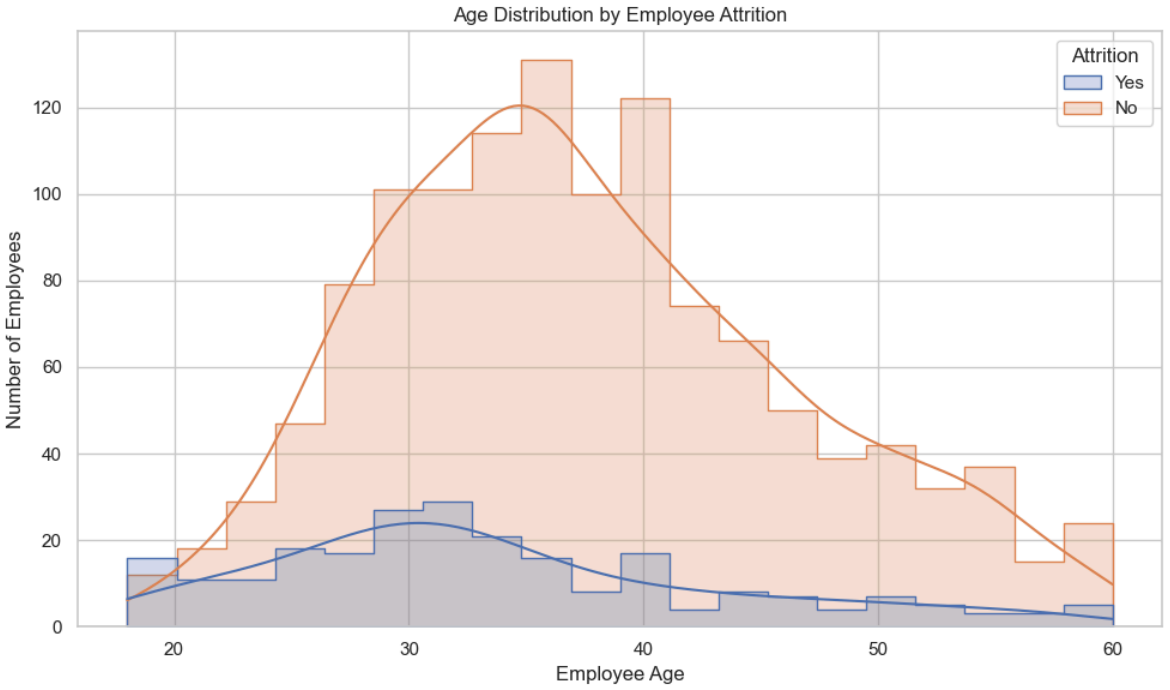


Analyze attrition by age

In [22]:
```
plt.figure(figsize=(10, 6))
```

```
sns.histplot(
    data=hr,
    x="Age",
    hue="Attrition",
    bins=20,
    kde=True,
    element="step"
)

plt.title("Age Distribution by Employee Attrition")
plt.xlabel("Employee Age")
plt.ylabel("Number of Employees")
plt.tight_layout()
plt.show()
```



In [23]:
```
age_analysis = hr.groupby("Attrition")["Age"].agg(
    ["count", "mean", "median"]
).round(2)

display(age_analysis)
```

Attrition	count	mean	median
No	1233	37.56	36.0
Yes	237	33.61	32.0

Analyze monthly salary

In [24]:
```
salary_analysis = hr.groupby("Attrition")["Monthly Income"].agg(
    ["count", "mean", "median", "min", "max"]
).round(2)

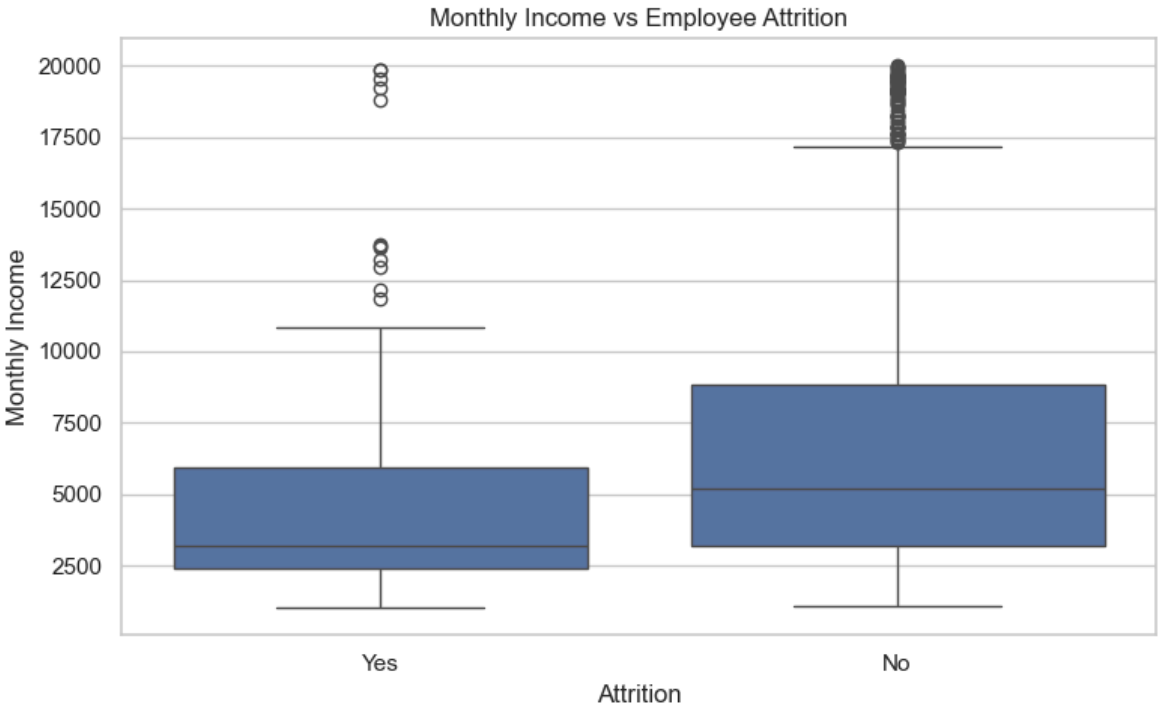
display(salary_analysis)
```

Attrition	count	mean	median	min	max
No	1233	6832.74	5204.0	1051	19999
Yes	237	4787.09	3202.0	1009	19859

In [25]:
```
plt.figure(figsize=(8, 5))

sns.boxplot(
    data=hr,
    x="Attrition",
    y="Monthly Income"
)

plt.title("Monthly Income vs Employee Attrition")
plt.xlabel("Attrition")
plt.ylabel("Monthly Income")
plt.tight_layout()
plt.show()
```



Analyze job satisfaction

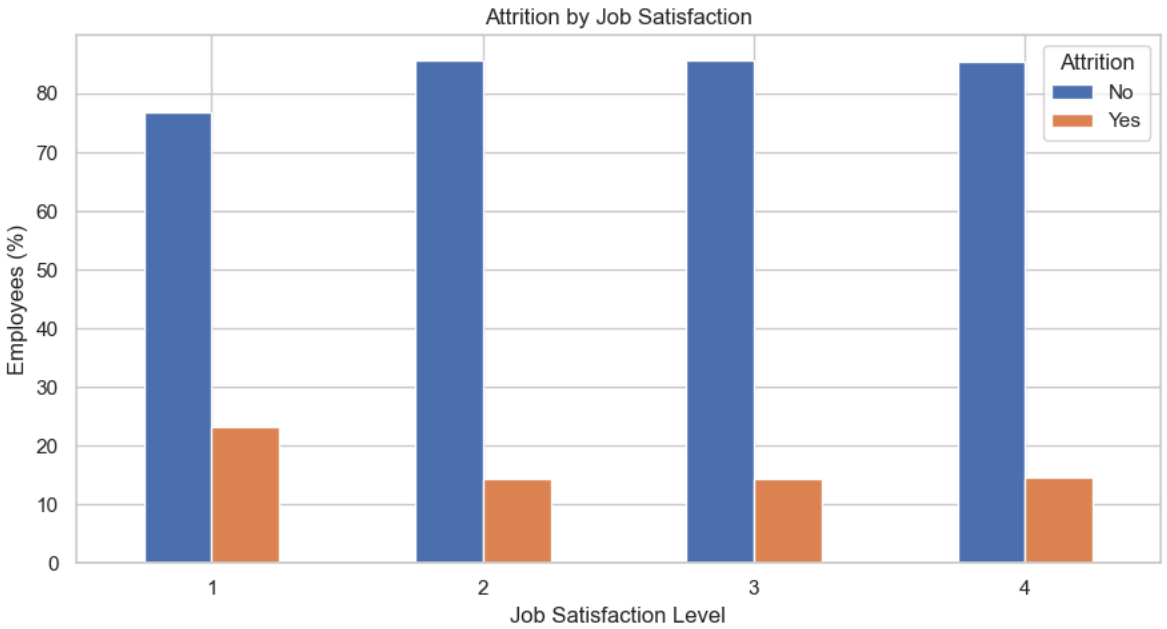
In [26]:
```
satisfaction_analysis = pd.crosstab(
    hr["Job Satisfaction"],
    hr["Attrition"],
    normalize="index"
).mul(100).round(2)

display(satisfaction_analysis)
```

Attrition	No	Yes
Job Satisfaction		
1	76.79	23.21
2	85.56	14.44
3	85.61	14.39
4	85.45	14.55

In [27]:
```
satisfaction_analysis.plot(
    kind="bar",
    figsize=(9, 5)
)

plt.title("Attrition by Job Satisfaction")
plt.xlabel("Job Satisfaction Level")
plt.ylabel("Employees (%)")
plt.xticks(rotation=0)
plt.legend(title="Attrition")
plt.tight_layout()
plt.show()
```



Analyze attrition by job role

In [28]:
```
role_analysis = hr.groupby("Job Role").agg(
    Total_Employees=("Attrition", "size"),
    Employees_Left=("Attrition", lambda x: (x == "Yes").sum())
)

role_analysis["Attrition_Rate"] = (
    role_analysis["Employees_Left"]
    / role_analysis["Total_Employees"] * 100
).round(2)

role_analysis = role_analysis.sort_values(
```

```
    "Attrition_Rate", ascending=False
)

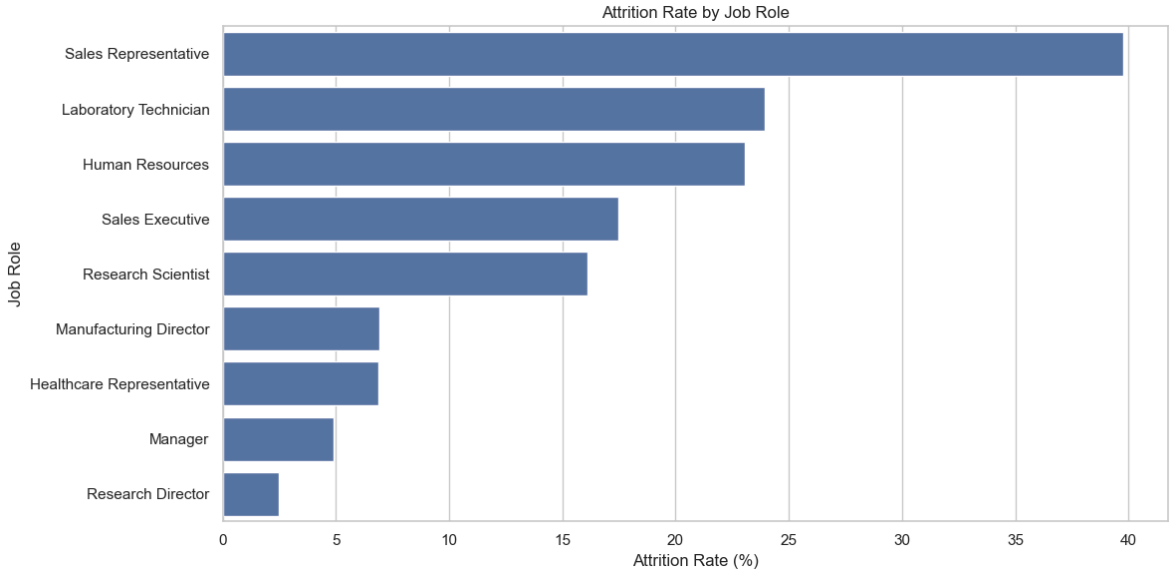
display(role_analysis)
```

Job Role	Total_Employees	Employees_Left	Attrition_Rate
Sales Representative	83	33	39.76
Laboratory Technician	259	62	23.94
Human Resources	52	12	23.08
Sales Executive	326	57	17.48
Research Scientist	292	47	16.10
Manufacturing Director	145	10	6.90
Healthcare Representative	131	9	6.87
Manager	102	5	4.90
Research Director	80	2	2.50

In [29]:
```
plt.figure(figsize=(12, 6))

sns.barplot(
    data=role_analysis.reset_index(),
    x="Attrition_Rate",
    y="Job Role"
)

plt.title("Attrition Rate by Job Role")
plt.xlabel("Attrition Rate (%)")
plt.ylabel("Job Role")
plt.tight_layout()
plt.show()
```



Use NumPy for numerical analysis

In [30]:
```
left_income = hr.loc[
    hr["Attrition"] == "Yes", "Monthly Income"
].to_numpy()

stayed_income = hr.loc[
    hr["Attrition"] == "No", "Monthly Income"
].to_numpy()

print("Average income of employees who left:",
      round(np.mean(left_income), 2))

print("Average income of employees who stayed:",
      round(np.mean(stayed_income), 2))

print("Median income of employees who left:",
      np.median(left_income))

print("Median income of employees who stayed:",
      np.median(stayed_income))
```

```
Average income of employees who left: 4787.09
Average income of employees who stayed: 6832.74
Median income of employees who left: 3202.0
Median income of employees who stayed: 5204.0
```

In [32]:
```
numeric_columns = [
    "Age",
    "Monthly Income",
    "Job Satisfaction",
    "Years At Company",
    "Years In Current Role",
    "Years With Curr Manager",
    "Distance From Home"
]

correlation = hr[numeric_columns].corr()

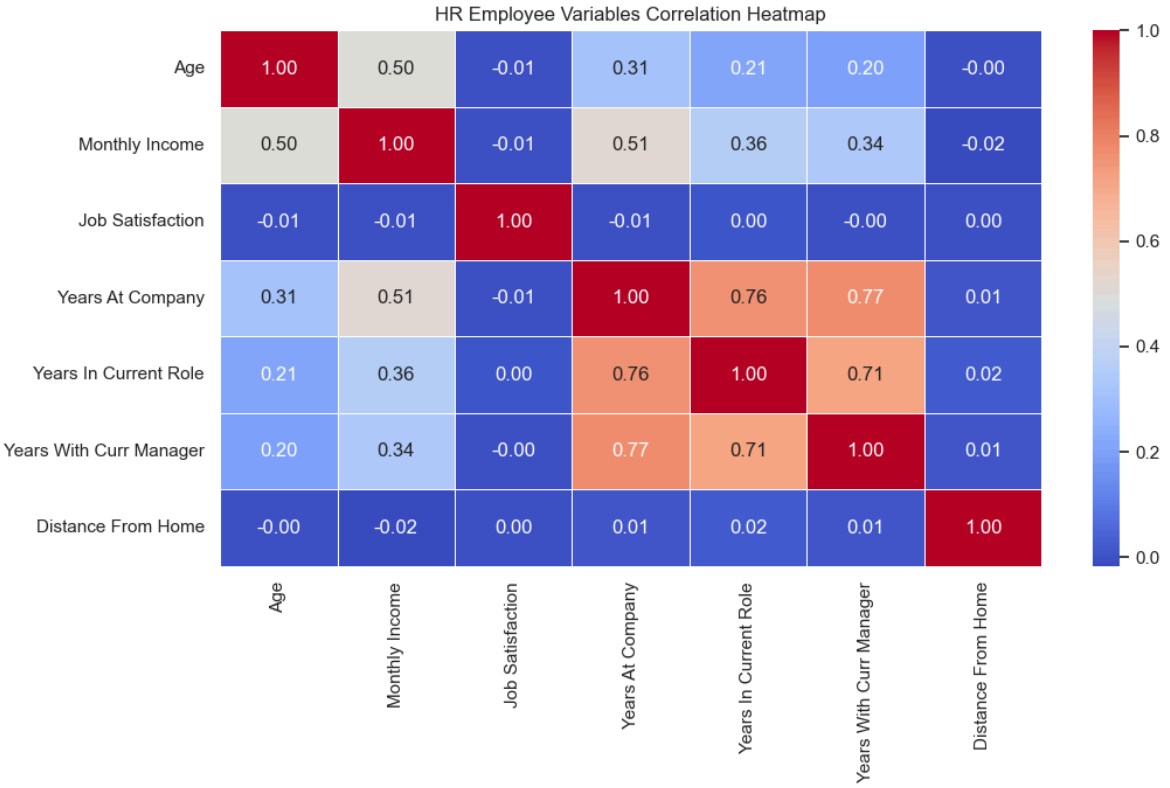
display(correlation.round(2))
```

	Age	Monthly Income	Job Satisfaction	Years At Company	Years In Current Role	Years With Curr Manager	Distance From Home
Age	1.00	0.50	-0.01	0.31	0.21	0.20	-0.00
Monthly Income	0.50	1.00	-0.01	0.51	0.36	0.34	-0.02
Job Satisfaction	-0.01	-0.01	1.00	-0.01	0.00	-0.00	0.00
Years At Company	0.31	0.51	-0.01	1.00	0.76	0.77	0.01
Years In Current Role	0.21	0.36	0.00	0.76	1.00	0.71	0.02
Years With Curr Manager	0.20	0.34	-0.00	0.77	0.71	1.00	0.01
Distance From Home	-0.00	-0.02	0.00	0.01	0.02	0.01	1.00

In [33]:
```
plt.figure(figsize=(11, 7))

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f",
    linewidths=0.5
)

plt.title("HR Employee Variables Correlation Heatmap")
plt.tight_layout()
plt.show()
```



Create an age-group analysis

In [34]:

```
conditions = [
    hr["Age"] < 25,
    hr["Age"].between(25, 34),
    hr["Age"].between(35, 44),
    hr["Age"].between(45, 54),
    hr["Age"] >= 55
]

choices = [
    "Under 25",
    "25-34",
    "35-44",
    "45-54",
    "55+"
]

hr["Age Group"] = np.select(
    conditions,
    choices,
    default="Unknown"
)

age_group_analysis = hr.groupby("Age Group").agg(
    Total_Employees=("Attrition", "size"),
    Employees_Left=("Attrition", lambda x: (x == "Yes").sum())
)

age_group_analysis["Attrition_Rate"] = (
    age_group_analysis["Employees_Left"]
    / age_group_analysis["Total_Employees"] * 100
).round(2)
```

```
display(age_group_analysis)
```

Age Group	Total_Employees	Employees_Left	Attrition_Rate
25-34	554	112	20.22
35-44	505	51	10.10
45-54	245	25	10.20
55+	69	11	15.94
Under 25	97	38	39.18

In [35]:
```
import os

os.makedirs("data", exist_ok=True)
os.makedirs("outputs", exist_ok=True)

hr.to_csv("data/cleaned_hr_data.csv", index=False)

print("Cleaned dataset saved successfully!")
```

```
Cleaned dataset saved successfully!
```

Generate a final HR summary report

In [36]:
```
total_employees = len(hr)
employees_left = (hr["Attrition"] == "Yes").sum()
employees_stayed = (hr["Attrition"] == "No").sum()

overall_attrition = np.mean(
    hr["Attrition"].eq("Yes")
) * 100

print("=" * 45)
print("       HR ATTRITION ANALYSIS REPORT")
print("=" * 45)

print("Total Employees:", total_employees)
print("Employees Left:", employees_left)
print("Employees Stayed:", employees_stayed)
print("Overall Attrition Rate:",
      round(overall_attrition, 2), "%")

print("\nDepartment Analysis:")
print(department_analysis.sort_values(
    "Attrition_Rate", ascending=False
))

print("\nJob Role Analysis:")
print(role_analysis)

print("\nOvertime Analysis:")
print(overtime_analysis)
```

```
print("\nSalary Analysis:")
print(salary_analysis)

print("=" * 45)
```

```
=============================================
       HR ATTRITION ANALYSIS REPORT
=============================================
Total Employees: 1470
Employees Left: 237
Employees Stayed: 1233
Overall Attrition Rate: 16.12 %

Department Analysis:
            Total_Employees  Employees_Left  Attrition_Rate
Department
Sales                   446              92           20.63
HR                       63              12           19.05
R&D                     961             133           13.84

Job Role Analysis:
                           Total_Employees  Employees_Left  Attrition_Rate
Job Role
Sales Representative                    83              33           39.76
Laboratory Technician                  259              62           23.94
Human Resources                         52              12           23.08
Sales Executive                        326              57           17.48
Research Scientist                     292              47           16.10
Manufacturing Director                 145              10            6.90
Healthcare Representative              131               9            6.87
Manager                                102               5            4.90
Research Director                       80               2            2.50

Overtime Analysis:
           Total_Employees  Employees_Left  Attrition_Rate
Over Time
No                    1054             110           10.44
Yes                    416             127           30.53

Salary Analysis:
           count     mean  median   min    max
Attrition
No          1233  6832.74  5204.0  1051  19999
Yes          237  4787.09  3202.0  1009  19859
=============================================
```

In [ ]:

In [ ]: